

The U.S. Military-Industrial Base: Competing in a Multipolar World

As the world enters an era of multipolarity, U.S. adversaries are closing the gap in military power. War games project the U.S. struggling to effectively compete in a protracted war with China as long-range missile stockpiles run out after just three weeks of conflict.¹ The American military-industrial base's response to empty stockpiles is a two-year delay before delivering the first replacement, limiting the U.S. military in a full-scale conflict for the future of the Indo-Pacific.¹ A shrinking defense-industrial base, fragile supply chains, and a declining K-12 education system pose a risk to decades of American military dominance. The U.S. maintains some advantages, but China, Russia, Iran, and North Korea now have the collective capability to cripple and overwhelm the U.S. military-industrial base. Despite lingering advantages, the emerging gaps in the U.S. military-industrial base and education system, paired with adversaries' rapid military advancements and deepened collaboration, threaten the U.S.' ability to deter and win a prolonged conflict.

Three decades after the unipolar moment, the United States still benefits from a technological edge, an unrivaled alliance network, and a world-class higher education system. These remaining advantages allow the U.S. to adapt and address increased competition abroad. The U.S.' technical advantage is built upon its advanced systems like nuclear submarines, interoperability with allies, and access to the world's most innovative private sector. The Department of Defense (DoD) is expanding its reach into the private sector, increasing long-term funding and investing in automation to mitigate the relative decline of American military power.² Outside its borders, the U.S. is leveraging its vast network of allies and partners to secure supply chains, reshore and friend-shore manufacturing capacity, and improve standardization for heightened interoperability.² The ongoing Russia-Ukraine War has expedited increases in allies' defense spending and improved Western collaboration on missile defense, cyber operations, and economic coercion to address the looming threat of an isolated Russia and rising China.³ At home, the U.S. continues to leverage STEM talent to innovate at unprecedented levels while maintaining the best universities in the world.⁴ Maintaining one of the best STEM-producing higher education systems will ensure that students from the U.S. and allies abroad are qualified and capable of supporting military needs. Additionally, burden-sharing with allies and investing in STEM talent will build upon these asymmetric advantages and reduce the risks of increased competition.

Despite these lingering advantages, the U.S. military-industrial base is still woefully unprepared for a prolonged conflict. The defense industrial base has contracted since 1985, resulting in a defense workforce reduction of 67% and 51 aerospace and defense prime contractors consolidating into just five companies.^{2 & 5} America's military now relies on three providers for over 90% of its missiles, while the number of contractors producing other critical capabilities, such as surface ships and satellites, decreased by 75% and 50%, respectively.^{5 & 6} While titans in the U.S. defense industry like Lockheed Martin and Raytheon build effective advanced weaponry and systems, they reduce competition for government funding and raise the

barrier of entry for startups to launch new, innovative products. A limited number of contractors impacts innovation, competition, and manufacturing capacity while leading to a one-size-fits-all approach to combat systems. As seen with the F-35, integrating capabilities into a single system risks cost overruns, raises design complexity, and complicates supply chains.^{2 & 7} Additionally, over-reliance on a handful of defense contractors allows adversarial actors to focus their cyber efforts and potentially infiltrate and exploit the majority of American defense technology.

Decades of globalization and off-shoring manufacturing created fragile and sporadic supply chains that support military systems. Defense contractors often rely on numerous international subcontractors to create advanced systems or weaponry. These complex supply chains leave the DoD blind to IP vulnerabilities and potential cost increases while limiting the U.S.' ability to surge capacity, respond to shock events, or prevent material shortfalls.^{2 & 5} Due to the resulting vetting process, the U.S. struggles to acquire technology competitively. Compared to China, the U.S. acquires cutting-edge technology from the private sector 20% as fast, even though it spends roughly \$310 billion of the defense budget on acquisitions annually.^{5 & 8} Relying on diverse supply chains may allow defense contractors to reduce costs, but they expose the American military to unnecessary delays and risks. For example, Taiwan and Japan play a crucial role in F-35 production, but American access to their products would be severely impacted in a conflict with China. Exposure to at-risk sources jeopardizes the U.S.' ability to react to national security threats.

The American K-12 education system no longer produces the STEM talent needed to support America's ambitions as a global superpower. With American students earning the lowest math scores in history, the U.S.' access to a skilled workforce and STEM talent is decreasing.⁹ To prevent decreases in productivity and revive domestic innovation, the DoD is incentivizing companies to upskill, reskill, and retain critical workers through the CHIPS and Science Act.² New development opportunities for academia, industry, and government workers to collaborate and learn more about strategic fields like robotics, semiconductors, and advanced materials are also emerging.² These new programs aim to produce enough STEM talent in the U.S. to fill military needs, such as hiring over 100,000 workers to meet submarine construction and sustainment needs in the next decade.² If the U.S. does not strengthen and develop STEM talent, the U.S.' technological advantage may prove to be unsustainable and allow adversaries to overtake the American military.

Russia and China have invested in manufacturing capabilities, while the U.S. industrial base has declined. The ongoing Russia-Ukraine conflict has led to massive investments in Russia's domestic manufacturing. Russia can now produce nearly 300% of the artillery munitions as the U.S. and Europe combined.¹⁰ Although Russia has mobilized its economy for war and holds the largest nuclear weapons stockpile, Chinese manufacturing capabilities pose a greater risk to the West. With a shipbuilding capacity that is 230 times larger than the U.S. and control over 70% of the global drone market, China's manufacturing capacity vastly outpaces the U.S. industrial base.^{8, 11, & 12} The Russia-Ukraine conflict has highlighted the critical need for drones in future warfare as each side leverages drones to destroy machinery, troops, and

infrastructure. China is already flexing its command of the industry as it ships just one drone to Ukraine for every 100 it sends to Russia.¹³ With the Ukrainian military estimating it needs 30,000 drones each month, the U.S.' inability to produce cheap drones at scale will be a significant disadvantage in conflicts against a China-backed alliance.¹³ North Korea and Iran further complicate the challenge facing the U.S. as they continue to make strides in their domestic missile development programs.³

In addition to China's robust manufacturing sector, the Chinese Communist Party (CCP) has prioritized developing resilient supply chains and controlling chokepoints. China controls 90% of global separation and refining processes for critical minerals and produces 75% of the world's critical minerals for semiconductor production.¹⁴ The CCP recently banned exports of refined germanium to the U.S., a critical mineral for which China controls 98.8% of global production.¹⁵ By dominating access to critical minerals and chokepoints in supply chains, China can surge manufacturing production while limiting adversaries' weapons development and energy production. China's civil-military fusion policies allow the CCP to quickly adapt private sector innovation to meet military needs. Even though the COVID-19 pandemic highlighted supply chain risks, private sector hesitation has slowed Western nations' ability to derisk and protect key sectors.

American adversaries have increasingly worked with one another to blunt the impact of aid to Ukraine and Western sanctions. Support from China, North Korea, and Iran propped up a teetering Russian war effort in Ukraine. North Korean troops and Iranian drones, alongside Chinese engines, missile technology, and microelectronics, are supporting Russia's war machine.¹⁰ This exchange of technology and supplies drives all U.S. adversaries' economies and militaries together. Iran and China have enhanced economic ties as Iran supplies China with 10% of its oil needs, and China plans to invest over \$400 billion in Iran.¹⁶ China's economic support also extends to North Korea, helping North Korea, Iran, and Russia evade Western sanctions. Iran and Russia now complete all transactions outside the Western-led SWIFT payment system and are constructing a drone factory in Russia.¹⁷ The Russia-Ukraine War cultivated an environment for U.S. adversaries to share military technology, deepen economic ties, and establish avenues to evade Western sanctions.

Russia, China, Iran, and North Korea are likely to continue their alliance of convenience into the future, increasing the level of U.S. military capabilities needed to deter or win a conflict. After decades of reducing domestic manufacturing, exposing supply chains to risk, and stagnant investment in education, modifying the U.S. military-industrial base poses a significant challenge. The U.S. must increase its manufacturing capacity, supply chain resilience, and K-12 education system effectiveness to strengthen the American military-industrial base and compete and win in a multipolar world.

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